Shiva Kar

Java Backend Developer | BCA Student

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SUMMARY

BCA Computer Science student (CGPA 8.84) specializing in Java backend development with Spring Boot, Hibernate/JPA, PostgreSQL, Redis and REST APIs. Built a Unix-like shell in C and has 6 merged pull requests in Godot Engine.

TECHNICAL SKILLS

Languages: Java, C, SQL

Java & Backend: Core Java, OOP, Collections, Generics, Multithreading, Spring Boot, Spring Framework, Spring MVC, Spring Data JPA, Hibernate, Spring Security, REST APIs, JWT, Dependency Injection, Transaction Management

Databases: PostgreSQL, MySQL, Redis, Database Design

Testing & Development Tools: JUnit 5, Mockito, Spring Boot Test, Integration Testing, Maven, Git, GitHub, Docker, Docker Compose, Flyway, OpenAPI

Computer Science & Systems: Data Structures and Algorithms, Operating Systems, Computer Networks, Linux, POSIX System Calls, Sockets, Memory Management, Concurrency

OPEN-SOURCE EXPERIENCE

Contributor, Godot Engine Aug 2026 – Present

github.com/godotengine/godot

- Submitted 15 pull requests (6 merged, 7 open, 2 closed) to the editor, GUI, GDScript and web platform. Key merged PRs: [PULL REQUEST URL], [PULL REQUEST URL].
- Fixed script tooltips appearing outside the script editor (#124170), "Save on Focus Loss" recreating deleted open scenes (#124931), resource reference counts not updating when resetting a property (#124789), integer zoom with fractional display scales (#124942) and a Web Editor file-deletion error (#123035).
- Revised a diagnostic tooltip feature (#123310) after review from four maintainers: moved row construction into a dedicated class and fixed hover-state bugs. Disabled text selection in diagnostic tooltips (#124054).

PROJECTS

E-Commerce Backend API | Java, Spring Boot, PostgreSQL **[PLANNED]** [MMM YYYY – MMM YYYY]

[GITHUB REPOSITORY URL]

- Designed REST APIs for products, categories, carts, orders and user accounts using a layered controller-service-repository architecture, DTO-based request validation, pagination, filtering and centralized exception handling.
- Secured endpoints with Spring Security, JWT authentication, password hashing and role-based authorization; modeled the PostgreSQL schema with Spring Data JPA/Hibernate, relational constraints and Flyway migrations.
- Implemented transactional order processing and inventory reservation with locking-based concurrency control to prevent overselling; tested with JUnit 5, Mockito and integration tests, and run with Docker Compose.

API Rate Limiter & Usage Metering Service | Java, Spring Boot, Redis **[PLANNED]** [MMM YYYY – MMM YYYY]

[GITHUB REPOSITORY URL]

- Built a Spring Boot service enforcing configurable per-user and per-API-key limits with a Token Bucket algorithm.
- Stored limiter state in Redis with TTL expiry and atomic Lua scripts to stay correct under concurrent requests.
- Returned HTTP 429 responses with Retry-After and rate-limit headers, recorded per-client usage metering, and tested boundary, expiry and concurrent-burst behavior with JUnit 5, Mockito and integration tests.

SXSHELL, Unix-like Shell | C, POSIX, CMake Mar 2026 – Aug 2026

github.com/shiva-kar/SXSHELL-unix-like-shell

- Built a modular shell in C (7 modules, about 550 lines) that separates command parsing from process execution.
- Implemented pipelines, I/O redirection, built-in commands and executable lookup using fork(), execv() and dup2().

EDUCATION

Bachelor of Computer Applications (BCA), Computer Science Jul 2025 – Expected Nov 2028

Amity University Online | CGPA: 8.84/10

Relevant Coursework: Object-Oriented Programming (Java), Data Structures using C, Operating Systems, Database Management Systems, Computer Networks, Unix and Shell Programming.

Class XII, Science | Deshabandhu Bidyapith H.S. School, Hojai, Assam, India | 68% 2022 – 2025

CERTIFICATIONS

CS50x: Introduction to Computer Science | Harvard University (CS50) [YEAR]

CS50 SQL: Introduction to Databases with SQL | Harvard University (CS50) [YEAR]

Languages: Bengali (Native), Hindi (Professional), English (Professional)